

REX REMIGIUS STEPHEN JOTHI

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EDUCATION

Northeastern University, Boston, MA

Master of Science in Computer Science

Sep 2025 - May 2027

(GPA: 3.93 / 4.00)

Relevant Coursework : Programming Design Paradigm, Computer Systems, Algorithms, Machine Learning, Operating System Implementation

Sri Krishna College of Engineering and Technology, Coimbatore, India

Bachelor of Technology in Information Technology

2019 - 2023

(GPA: 8.68 / 10.00)

SKILLS

Programming Languages:

C, C++, Python, Java, Bash

Compiler & Systems:

LLVM, GCC, Intrinsic Programming, SIMD Optimization

AI/ML Performance:

Model Quantization, ONNX, TensorFlow Lite, PyTorch, Kernel Optimization, Roofline Modeling, Benchmarking, Profiling

Development & Testing Tools:

Linux, Git, CMake, Make, Docker, GDB, Pytest, LLVM LIT, perf

Hardware:

RISC-V, Qualcomm Snapdragon NPU, Cadence Vision P6 DSP (Xtensa), ARM

WORK EXPERIENCE

AI Research Intern, Qualcomm Technologies Inc, San Diego, CA

Jun 2026 - Sep 2026

- Built an agent-assisted performance-analysis workflow that detects kernel-level bottlenecks from profiling artifacts, execution traces, and runtime logs for AI workloads on Qualcomm mobile AI accelerators.
- Built calibrated roofline performance models for Qualcomm AI hardware to characterize and optimize LLM and VLM inference workloads.
- Reduced performance-debugging turnaround from days to under a few hours by automating analysis and bottleneck identification.

Software Engineer, MulticoreWare Inc, Chennai, India

Jan 2023 - Jul 2025

- Designed custom intrinsics and optimization passes in LLVM and GCC, enabling reusable target-specific code and improving generated code efficiency by 15% through memory-aware transformations.
- Architected and optimized ML kernels (Convolution, Maxpool, Normalization) on a custom AI accelerator using SIMD and cache-efficient memory layouts, cutting latency 63% single-stream and 67% multi-stream.
- Ported ML workloads from ARM CPU to Cadence Vision P6 DSP via the Xtensa simulator, using memory modeling to validate theoretical estimates and guide kernel optimization.
- Converted PyTorch models to ONNX and fully quantized TFLite, with automated verification for bit-exact output consistency and low-latency inference across platforms.

PROJECTS

Enhanced Address Sanitizer pass in LLVM for fault tolerance

C, C++, LLVM, Shell, Make

- Extended LLVM's AddressSanitizer to continue fault-tolerant execution instead of aborting on memory errors, implementing the CIMA (Countering Illegal Memory Access) algorithm with CFG-based instrumentation.
- Integrated custom LLVM/Clang passes, enhanced reporting, and adaptive memory resizing to reach 100% runtime availability, 0% crash rate, and 30% lower reallocation overhead, validated with IR-level testing and C benchmarks.

Transparent Checkpoint and Restart System

C, POSIX signals, Linux syscalls, GDB

- Built a transparent checkpoint-restart mechanism capturing full process state (memory segments, CPU register contexts) via /proc/self/maps and getcontext/setcontext to restore execution across new address spaces without program modifications.
- Implemented state validation protocols through controlled restarts, ensuring memory segment integrity and consistent runtime register restoration.

ReceiptVLM

Python, PyTorch, MLX, Hugging Face, QLoRA, FastAPI

- Fine-tuned Qwen2.5-VL-3B with QLoRA on WildReceipt to emit structured JSON directly from receipt images, reaching 0.781 micro-F1 (3.7x zero-shot, 6.8x a Tesseract+regex baseline) with 47% lower latency and 50% lower memory via INT4 quantization, validated with bootstrap CIs and paired significance tests.
- Ported the MLX training and inference stack to Hugging Face transformers/PEFT, diagnosing a 0.785 to 0.288 accuracy collapse on adapter transfer as base-model mismatch, and restored parity by re-fitting on the HF base with fp16 serving (+0.038 micro-F1 over NF4).